

Incident Investigation Report: Command Shell Execution on DESKTOP-5OG7C28

Oct 7, 2026 · @Darragh

1. Executive Summary

Between 14:52:59 and 14:55:27 UTC on 2 October 2026, an elevated session belonging to local user **DESKTOP-5OG7C28\Darragh Casey** executed a batch script, `C:\AtomicRedTeam\ExternalPayloads\T1059.003_script.bat`. The script was launched through PowerShell (`Start-Process`), which spawned `cmd.exe /c`. The file paths and working directory name an adversary-emulation framework and the ATT&CK technique ID T1059.003. This indicates a deliberate, controlled test of Windows Command Shell execution, consistent with the isolated lab context. It does not indicate a genuine intrusion.

The activity occurred in two invocations about 2.5 minutes apart. The first ran only a file-existence check (`Test-Path`) on the script. The second repeated that check and then executed the script. Each PowerShell command was preceded by `hostname.exe` and `whoami.exe` executions from the same unresolved parent process (PID 4408).

The three highest-severity alerts are **false positives**. These are Wazuh rule 92213 (level 15, "Executable file dropped in folder commonly used by malware", mapped to T1105). Each was triggered by PowerShell's own `__PSScriptPolicyTest_*.ps1` temporary files, created within about 160 ms of each PowerShell process starting. Nothing was transferred into the host. The behaviour that actually reflects the tested technique, PowerShell spawning `cmd.exe` (rule 92004), was alerted at only level 4.

The telemetry shows nothing the script did after `cmd.exe` started. The Sysmon record sequence is contiguous through 14:59:42, and no further Sysmon events attributable to the script were recorded. The remaining events are routine Windows background activity: Edge/OneDrive updaters, Task Scheduler tasks, SmartScreen, the Update orchestrator, and BITS start-type changes.

Overall confidence: **high** for the execution chain and the false-positive assessment, **medium** for the interpretation of the `hostname / whoami` activity, and **low** for any explanation of the BITS changes.

2. Scope of Activity

The supplied data covers one Wazuh agent (001 , Windows_VM , 192.168.232.130), reporting host DESKTOP-5OG7C28 to manager dcasey-virtual-machine . The window runs from 14:50:40 to 14:59:42 UTC (16:50–16:59 local, UTC+02:00).

The data comprises:

- Sysmon events with record IDs 15228–15252, a contiguous sequence with no gaps.
- Four Security log events (54205–54208, contiguous).
- Two System log events (3018–3019).
- Six Wazuh alerts.

Every alert in the alerts file also appears, with identical rule metadata, in the archive file. The archive is therefore the authoritative and complete dataset.

The relevant activity is confined to:

- One host and one user account.
- One elevated logon session (LogonId 0xA8D6D , integrity level High).
- One parent process (PID 4408).
- One script file.

There is no evidence of network activity, lateral movement, persistence, credential access, or privilege escalation associated with the test activity. The only outbound network connection in the window belongs to the OneDrive updater, which is unrelated.

Two points about timestamps. Wazuh timestamp values are ingestion times, roughly 0.5–1 s after the event. The archive file is also not strictly chronological: the whoami record 15235 appears after PowerShell record 15236. This report orders events by the Windows systemTime / UtcTime field and the event record ID.

3. Detailed Investigation

3.1 The execution chain (observed)

The central event is Sysmon record 15251 (Event ID 1) at 14:55:27.411 UTC:

- **Process:** C:\Windows\System32\cmd.exe (PID 8188).
- **Command line:** C:\WINDOWS\system32\cmd.exe /c
 ""C:\AtomicRedTeam\ExternalPayloads\T1059.003_script.bat" "

- **Session:** user Darragh Casey, integrity High, LogonId 0xA8D6D .
- **Parent:** powershell.exe (PID 1212, ProcessGuid {4b2ce92d-c5de-6abf-7e04-...}).
- **Parent command line:** "powershell.exe" & {Start-Process
\\\"C:\AtomicRedTeam\atomics\..\ExternalPayloads\T1059.003_script.bat\"\"}

The parent's path atomics\..\ExternalPayloads normalises to the same ExternalPayloads folder that appears in the child command line, so both refer to one file. PowerShell PID 1212 was itself created at 14:55:26.958 (record 15248) by PID 4408.

The parent-child relationship is fully attested by matching ProcessGuid values. This is a direct, observed PowerShell → cmd.exe → batch script execution chain.

3.2 Two invocations (observed sequence, inferred purpose)

The same pattern appears three times, always from parent PID 4408 in the same elevated session: hostname.exe , then whoami.exe , then a powershell.exe command.

Invocation 1 (14:52:59–14:53:00):

1. hostname (PID 2576).
2. whoami (PID 8576).
3. powershell PID 7164 running if (Test-Path \"..\T1059.003_script.bat\") {exit 0} else {exit 1} .

Nothing else followed. *Inference:* this was a prerequisite check only. The script was not executed in this invocation.

Invocation 2 (14:55:25–14:55:27):

1. Three hostname executions (PIDs 10356, 10696, 10808).
2. whoami (10384).
3. The same Test-Path check (PowerShell PID 4484).
4. A further hostname (1712) and whoami (10400).
5. powershell PID 1212 running Start-Process on the script.
6. cmd.exe launching the script.

Inference: this was a prerequisite check followed by execution.

3.3 Interpretation of hostname/whoami (inferred, medium confidence)

Running `hostname` and `whoami` could in principle represent System Information Discovery and System Owner/User Discovery. However, several features of the evidence point to harness overhead rather than deliberate discovery:

- **Fixed order and timing.** The pairs occur in an unchanging order and immediately precede each PowerShell command, by roughly 240–280 ms.
- **Different working directories.** The pairs run with working directory `C:\Users\Darragh Casey\Desktop\T1059.003\`. The test commands run with `C:\Users\DARRAG-1\AppData\Local\Temp\`. This suggests the orchestrating process set a separate working directory for test commands.
- **Script-block launch syntax.** The `"powershell.exe" & {...}` command-line form indicates a script block passed by an orchestrating process.

The more likely explanation is that these calls are the orchestrating tool recording host and user context for each execution step, rather than discovery activity in their own right. This cannot be proven from the telemetry. Both interpretations are reflected in the ATT&CK mapping with appropriate caveats.

3.4 The parent process PID 4408 (observed gap)

For PID 4408, Sysmon reports `ParentProcessGuid {00000000-...}` and `ParentImage -`, so the image of the orchestrating process cannot be determined from the supplied data. The same happens for PID 952, the parent of COM-activated `SoftLandingTask.exe` and `smartscreen.exe`. Sysmon resolved other long-running parents normally, such as the Task Scheduler `svchost` (PID 1268), so the reason these two are unresolved cannot be determined here.

Inference: PID 4408 is a PowerShell-based orchestration session. This rests on the script-block syntax of its children and the framework paths involved, but it is not directly observed.

3.5 Session context (observed and inferred)

Two logon IDs for the same user appear in the data:

- **0xA8D6D , integrity High.** All test-related processes ran here.
- **0xA8DA9 , integrity Medium.** Background user processes ran here: the OneDrive updater, `taskhostw`, and `smartscreen`.

Their LogonGuids are adjacent (`...6d8d...` and `...a98d...`). *Inference:* this is a UAC split-token administrator, and the test was deliberately run from an elevated ("Run as administrator") session. Execution at High integrity is meaningful because any commands in the batch script would have inherited administrative rights.

3.6 SmartScreen (inferred)

`smartscreen.exe -Embedding` (PID 324) started at 14:55:27.343. That is 385 ms after PowerShell 1212 started and 68 ms before `cmd.exe` . *Inference:* the `Start-Process` (ShellExecute) launch of the `.bat` file triggered a SmartScreen reputation check. This is consistent with Windows behaviour, and the timing is tight, but no process relationship links them directly.

3.7 What the script did (not observable)

The script's contents and effects are not in the evidence. Several facts bear on this:

- The Sysmon record IDs run contiguously from 15228 to 15252.
- The only record after the `cmd.exe` launch (15251) is an unrelated Windows Update worker at 14:59:42 (15252).
- So within this Sysmon configuration, `cmd.exe` PID 8188 produced no logged child processes, file creations, network connections, or DNS queries in the following four minutes.

Two explanations fit: the script used only `cmd` built-ins, which create no new processes, or its actions fell outside the Sysmon configuration's filters. The configuration evidently filters selectively. For example, no process-termination events were logged for the short-lived `hostname` / `whoami` / `powershell` processes, yet one was logged for the OneDrive updater.

The script's own creation is also absent. *Inference:* it existed before 14:50:40, or its creation was not covered by the file-creation filter. The script certainly existed at 14:55:27, because `Start-Process` would have failed without launching `cmd.exe` if the file were missing.

3.8 Alert assessment

Rule 92213 (level 15, T1105), three alerts: false positive.

Record	Creating PowerShell PID	Process start (UTC)	File created (UTC)	Delta
15237	7164	14:53:00.058	14:53:00.175	117 ms
15245	4484	14:55:26.328	14:55:26.443	115 ms
15249	1212	14:55:26.958	14:55:27.119	161 ms

The created files are:

- `__PSScriptPolicyTest_jarjq0wa.5fg.ps1`
- `__PSScriptPolicyTest_z4baxwgm.atc.ps1`
- `__PSScriptPolicyTest_sct5nwuc.3fq.ps1`

All are in the user's Temp folder. Their randomised names follow a fixed prefix, and each was written by the newly started `powershell.exe` within milliseconds of launch. This is the pattern of PowerShell's start-up script-execution-policy test, not of a dropped payload.

Several further points support the false-positive assessment:

- The files are `.ps1` scripts, not PE executables.
- No network connection or DNS query was made by any PowerShell process.
- The T1105 mapping (Ingress Tool Transfer) is therefore unsupported.

These files are an indirect by-product of the test, since the test launched the PowerShell processes, but they are not malicious artifacts. The `firedtimes` values (11, 12, 13) show the rule had already fired ten times on this manager before this window, which suggests a recurring noise source.

Rule 92004 (level 4, T1059.003): true positive for the behaviour. PowerShell did spawn a Windows command shell to run a batch script. In context the activity is authorised emulation, not malicious, but the detection correctly identified the technique being exercised. The severity is inverted: the real technique alerted at level 4, while benign artifacts alerted at level 15 with email notification enabled.

Rule 61104 (level 3), two alerts: benign or unattributed. The BITS service start type changed from demand to auto at 14:51:13.681 (System 3018) and back to demand at 14:53:18.994 (System 3019). The event's processID is 812, which matches `services.exe` (0x32c) in the 4624 event at 14:51:13.568, so the Service Control Manager recorded the change.

No initiating process or account appears in the data. The first change came 1 min 46 s before any test activity. No BITS client activity was observed: no `bitsadmin`, no BITS-related process creation, and no transfers. The reversal at 14:53:18 falls 19 s after Invocation 1's `Test-Path`, but no causal link is evident. *Inference (low confidence)*: this is routine Windows servicing behaviour, unrelated to the test.

3.9 Excluded activity

The following events were reviewed and judged unrelated. They come from separate parent chains, run under SYSTEM or the medium-integrity token, and involve known Windows components.

- **Edge updater.** `MicrosoftEdgeUpdate.exe /ua /installsource core` (SYSTEM, parent Edge updater).
- **OneDrive updater.** `OneDriveStandaloneUpdater.exe /reporting` was launched by Task Scheduler (`svchost -s Schedule`). It resolved `ecs.office.com` to 52.123.128.14 / 52.123.129.14, then connected to 52.123.128.14:443. The destination matches the DNS answer.
- **svchost PID 1860 (0x744).** It resolved `oneclient.sfx.ms` to an Akamai CDN address. It also enumerated the Administrators and Backup Operators groups (4799) as SYSTEM. Neither event links to the test session.
- **Scheduled-task and COM activity.** `taskhostw.exe` and `SoftLandingTask.exe` (Windows tips/notification content).
- **Windows Update.** `MoUsoCoreWorker.exe` (SYSTEM).
- **Service logon.** The 4624 type-5 / 4672 SYSTEM service logon pair.

Excluding these does not change the conclusion. None share a process lineage, logon session, or file artifact with the test chain.

4. Evidence Analysis

Finding	Evidence	Type
Batch script executed via cmd.exe	Sysmon 15251: <code>cmd.exe</code> PID 8188 <code>/c</code> <code>"\"C:\AtomicRedTeam\ExternalPayloads\T1059.003_script.bat" "; Wazuh alert 92004</code>	Observed

Finding	Evidence	Type
PowerShell launched cmd.exe	15251 ParentProcessGuid {...7e04...} = 15248 ProcessGuid; PID 1212; Start-Process command line	Observed
Prerequisite checks before execution	15236 (PID 7164) and 15244 (PID 4484): Test-Path on the script, exit 0/1	Observed; purpose inferred
Single orchestrating parent	ParentProcessId 4408 on records 15234–15236 and 15240–15248 (excluding EID 11)	Observed; image unknown
Elevated execution	IntegrityLevel High, LogonId 0xA8D6D on all test processes; background tasks at Medium, 0xA8DA9	Observed; split-token inferred
92213 alerts are false positives	15237/15245/15249: __PSScriptPolicyTest_*.ps1 created by each new PowerShell within 115–161 ms of start; no network events from PowerShell	Observed facts; FP judgement inferred (high confidence)
No observable post-execution activity	Contiguous Sysmon IDs 15228–15252; next event after 15251 is unrelated MoUsoCoreWorker at 14:59:42	Observed; config-dependent
BITS changes unattributed	System 3018/3019 logged by services.exe (PID 812); no initiating process; first change precedes test	Observed; benign interpretation inferred (low)
Framework-driven test	Paths C:\AtomicRedTeam\..., folder Desktop\T1059.003, file T1059.003_script.bat	Observed strings; intent inferred

Hashes recorded for reference (all Microsoft-signed system binaries, per the Company/Product fields):

- **cmd.exe:** SHA256
97AC98B1A92C286054CCE55239CFCCDFC23A5517BD07FE693072C9CA96C7DABB
- **powershell.exe:** SHA256
8BB6FA8C283B4D92120B1EF249A9B311B0F804D4CABBE9981159976C8BE76A5E
- **hostname.exe:** SHA256
22EC2260B5C32177F7D14A33975E1005988BBA0DF3427CC8B47CEC671D3F2A1F

- **whoami.exe: SHA256**

23240EF9F8B0A9A324110B1C2331DE31DC1B0E08F5359CB707E51A939AF56CD3

No hash of the batch script itself is available.

5. Timeline of Events

All times are UTC on 2 October 2026; local time is UTC+2.

Time (UTC)	Record	Event	Relevance
14:50:40.624	Sysmon 15228	MicrosoftEdgeUpdate.exe (6004, SYSTEM) started	Background
14:51:13.330	Sysmon 15229	OneDriveStandaloneUpdater.exe (9192) started by Task Scheduler	Background
14:51:13.568	Security 54205/54206	4624 type 5 + 4672, SYSTEM via services.exe	Routine service logon
14:51:13.681	System 3018	BITS demand → auto (alert 61104)	Unattributed
14:51:17.942	Security 54207/54208	4799 Administrators / Backup Operators enumerated by svchost 1860	Background
14:51:19.176-.888	Sysmon 15230-15232	DNS (oneclient.sfx.ms, ecs.office.com) and TCP to 52.123.128.14:443	Background (OneDrive)
14:52:29.556	Sysmon 15233	OneDrive updater terminated	Background
14:52:59.793	Sysmon 15234	hostname.exe (2576), parent 4408, High	Invocation 1
14:52:59.820	Sysmon 15235	whoami.exe (8576), parent 4408	Invocation 1
14:53:00.058	Sysmon 15236	powershell (7164) Test-Path on script	Prereq check
14:53:00.175	Sysmon 15237	__PSScriptPolicyTest_jarjq0wa.5fg.ps1 (alert 92213)	False positive
14:53:18.994	System 3019	BITS auto → demand (alert 61104)	Unattributed
14:55:01.333-.392	Sysmon 15238-15239	taskhostw.exe; SoftLandingTask.exe	Background
14:55:25.945-26.060	Sysmon 15240-15242	hostname.exe ×3 (10356, 10696, 10808)	Invocation 2
14:55:26.091	Sysmon 15243	whoami.exe (10384)	Invocation 2
14:55:26.328	Sysmon 15244	powershell (4484) Test-Path on script	Prereq check

Time (UTC)	Record	Event	Relevance
14:55:26.443	Sysmon 15245	__PSScriptPolicyTest_z4baxwgm.atc.ps1 (alert 92213)	False positive
14:55:26.679-.710	Sysmon 15246-15247	hostname.exe (1712), whoami.exe (10400)	Invocation 2
14:55:26.958	Sysmon 15248	powershell (1212) Start-Process on script	Execution
14:55:27.119	Sysmon 15249	__PSScriptPolicyTest_sct5nwuc.3fq.ps1 (alert 92213)	False positive
14:55:27.343	Sysmon 15250	smartscreen.exe -Embedding (Medium)	Likely triggered by launch
14:55:27.411	Sysmon 15251	cmd.exe (8188) /c T1059.003_script.bat, parent 1212 (alert 92004)	Technique executed
14:59:42.223	Sysmon 15252	MoUsoCoreWorker.exe (SYSTEM)	Background; end of data

6. MITRE ATT&CK Mapping

Tactic	Technique	Supporting evidence	Confidence
Execution	T1059.003 Windows Command Shell	cmd.exe /c executing T1059.003_script.bat (15251); alert 92004	High
Execution	T1059.001 PowerShell	powershell.exe script blocks used for the prerequisite checks and to launch the script via Start-Process (15236, 15244, 15248)	High (as launcher)
Discovery	T1033 System Owner/User Discovery	whoami.exe ×3 (15235, 15243, 15247)	Observed. Medium that it reflects harness logging rather than discovery intent
Discovery	T1082 System Information Discovery	hostname.exe ×5 (15234, 15240-15242, 15246)	Observed. Same caveat as T1033

Mappings considered and rejected:

- **T1105 Ingress Tool Transfer**, as assigned by rule 92213. No transfer is evidenced: the files are PowerShell's own policy-test scripts, and no network activity came from the test processes.
- **T1197 BITS Jobs**. Only start-type changes were logged, with no BITS job creation or transfer, and nothing links them to the test.
- **T1543.003 / service modification**. The start-type changes have no attributable actor.
- **T1069.001 Local Groups discovery**, from the 4799 events. These were performed by a SYSTEM svchost, unrelated to the test session.

7. Overall Investigation Conclusion

Observed facts:

- In an elevated session of local user Darragh Casey on DESKTOP-5OG7C28, a process with PID 4408 ran two invocations.
- Both invocations checked for `C:\AtomicRedTeam\ExternalPayloads\T1059.003_script.bat`.
- The second then launched the script through PowerShell `Start-Process`, producing `cmd.exe /c` at 14:55:27.411 UTC.
- Each step was preceded by `hostname / whoami` executions.
- Three level-15 alerts fired on PowerShell's start-up policy-test files.
- One level-4 alert fired on the PowerShell → cmd.exe spawn.
- No further Sysmon activity attributable to the script was recorded, and no network, persistence, or credential activity was linked to the session.

Inferences:

- This was a deliberate, authorised adversary-emulation test of T1059.003, run from a PowerShell-based framework. This is supported by the framework and technique names in the paths and by the isolated lab context.
- The first invocation was a prerequisite check only.
- The `hostname / whoami` executions are most likely framework logging overhead.
- The level-15 alerts are false positives.
- The BITS and other background events are unrelated.

Significance: As a detection exercise, the outcome is mixed. The technique was detected, but the true-positive detection (92004) was low severity, while benign PowerShell artifacts generated the highest-severity, email-enabled alerts. In a production setting this would mislead triage toward a non-existent ingress transfer.

Recommendations:

- Tune rule 92213 to exclude `__PSScriptPolicyTest_*.ps1` written by `powershell.exe`.
- Review the severity of 92004 in context, for example by correlating it with elevated integrity or scripts in non-standard paths.
- Enable PowerShell Script Block Logging (4104) and capture `.bat` file creation, so script contents and effects become visible.
- Retrieve the earlier process-creation event for PID 4408 to close the parent-attribution gap.

Overall confidence:

- **High** that the evidence shows a controlled command-shell execution test with no malicious outcome observed, and that the 92213 alerts are false positives.
- **Medium** on the role of `hostname / whoami` and on the identity of PID 4408.
- **Low** on the cause of the BITS start-type changes.

The script's actual behaviour cannot be determined from the supplied telemetry.